

Project Report: Microelectronic Devices and Circuits

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Topic: Theoretical Analysis and Numerical Simulation of Semiconductors and Junctions

Text Reference: Microelectronic Devices and Circuits by Clifton G. Fonstad

Introduction

The objective of this assignment was to apply the fundamental semiconductor equations derived in the first six chapters of Fonstad to a 1D numerical solver using devsim software. Using a $50\mu\text{m}$ mesh with 100 nodes, I simulated two structures: a uniformly doped resistor and an p-n junction. The main goal was to observe how carrier concentrations (n,p) and potential will respond to external biases (0.3V for the resistor and $\pm 0.1\text{V}, 0.3\text{V}, 0.5\text{V}$ for the diode).

Theoretical Framework

The simulation is made using the theoretical logic from Microelectronic Devices and Circuits by Clifton G. Fonstad. The important theories used are:

- **Thermal Equilibrium :** The solver assumes that in equilibrium, the product of electron and hole concentrations is constant: $n_0 p_0 = n_i^2$. For this project, n_i was taken as 10^{10}cm^{-3} .
- **Transport Mechanisms :** Total current density is modeled as the sum of drift (due to electric fields) and diffusion (due to concentration gradients)
- **The Five Basic Equations :** numerical solver approximates the solution to Poisson's equation and the continuity equations given in the textbook to find the potential $\phi(x)$ and carrier densities across the $50\mu\text{m}$ mesh.

- **Flow Problems :** The simulation of the diode under bias is based on the theory of low level injection, where minority carrier flow is dominated by diffusion .

Implementation and Results of the project:

The Resistor (Uniform Doping)

I created a 1D mesh with 100 node points. I defined a net doping profile ($N_D - N_A$) using the constants c_1 and c_2 .

- **Observations:** At zero bias, the potential was found to be constant across the mesh, and excess carrier concentrations were zero. This confirms the equilibrium theory in the text book. This observation was made from the graph from paraview produced using devsim software
- **Applied Bias (0.3V):** With 0.3V applied, the potential shows a linear gradient. Because the semiconductor is uniform, the electric field is constant, and the current is purely drift-driven.

The p-n Junction Diode

I implemented a junction at $x=25\mu\text{m}$ with $N_D = 10^{17}$ and $N_A = 10^{17}\text{cm}^{-3}$.

- **Zero Bias (Equilibrium):** The solver successfully plotted the built in potential . According to depletion approximation given in the text book $\phi_b = (kt/q) \ln(N_A * N_D / n_i * n_i)$

With the given doping, the simulation showed a potential barrier of approximately 0.84V.

- **Biasing (Forward and Reverse):**
- **Forward Bias (0.1V, 0.3V, 0.5V):** The potential barrier was reduced. My plots obtained using devsim software showed a significant increase in excess carrier concentration at the junction edges, which then decayed as carriers moved into the quasineutral regions

- **Reverse Bias (-0.1V, -0.3V, -0.5V):** The depletion region width (w) visibly widened in the plots. This observation is in accordance with the theory of the Fonstad textbook .

Bonus part of assignment

The syntax was obtained for part 6 which focused on the current values from the contacts using documentation and for part 7 the voltage bias was changed from -0.7 to 0.7 . This results of this observation showed the transition from a reverse bias state to a forward biased state. The error limit of devsim software was also changed to solve convergence issue

Conclusion

The project helped me to learn a lot about the working of semiconductor devices like resistors and pn junction diode . It also provided a transition from the theoretical knowledge gained from the chapters of Fonstad to a functional numerical device set up using devsim software in python environment . It also provided me with the validation of the five basic equations and the dynamics of carrier injection.